

Lab Assignment 5

Session tracking with HttpSession

Aim

To develop web applications using Java Servlets that demonstrate session tracking using HttpSession for storing and retrieving user data across multiple requests.

Learning Objectives

1. Understand the concept of session tracking.
2. Learn how to use HttpSession object.
3. Store and retrieve session data using:
 - setAttribute()
 - getAttribute()
4. Develop multi-page applications using servlets.
5. Maintain user state across multiple requests.

Software Requirements

- Java JDK (8 or above)
- Apache Tomcat Server (9+)
- IDE (Eclipse / IntelliJ)
- Web Browser (Chrome)

5.1 Write a Servlet that accepts a User Name from a HTML form and stores it as a session variable. Write another Servlet that returns the value of this session variable and displays it.

Code:

getData.java

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;
```

```
@WebServlet("/getData")
```

```
public class getData extends HttpServlet {
    private static final long serialVersionUID = 1L;

    protected void doGet(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException {
        String name = request.getParameter("username");
        HttpSession session = request.getSession();
        session.setAttribute("username", name);
        response.setContentType("text/html");
        PrintWriter p = response.getWriter();
        p.println("<p>Session Created, Username Stored"
            + "<br>"
            + "<a href='displaySession'>Display Session</a>");
    }

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {
        doGet(request, response);
    }
}
```

displaySession.java

```
import java.io.IOException;
import java.io.PrintWriter;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

@WebServlet("/displaySession")
public class displaySession extends HttpServlet {
    private static final long serialVersionUID = 1L;

    protected void doGet(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException {
```

```
        HttpSession session = request.getSession();
        String name = (String) session.getAttribute("username");
        response.setContentType("text/html");
        PrintWriter p = response.getWriter();
        p.println("<h1>Welcome "+name+"</h1>");
    }

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {
        doGet(request, response);
    }
}
```

Login.html

```
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body>
<form action="getData" method="get">
Name: <input type="text" name = "username"><br><br>
<input type="submit" value="Submit">
</form>
</body>
</html>
```

Output:

Name:

Session Created, Username Stored
[Display Session](#)

Welcome Shriyash

5.2. Create a project for Raj Spa as follows:

A HTML form should display the services(face Spa,Body Spa,Foot Spa) with fixed charges in the form of checkboxes. On click of submit button these services should be stored in a session variable and a next HTML form should display the discount options(None,5%,10% and 15%) which would be stored in session variable. The last page should display the services selected and their charges, the discount and net payable.Use servlet.

Code:

Service.html

```
<!DOCTYPE html>
<html>
<head>
<title>Spa Services</title>
</head>

<body>

<h2>Select Spa Services</h2>

<form action="ServiceServlet" method="post">

<input type="checkbox" name="service" value="Face Spa"> Face Spa (₹500)<br>

<input type="checkbox" name="service" value="Body Spa"> Body Spa (₹1000)<br>

<input type="checkbox" name="service" value="Foot Spa"> Foot Spa (₹300)<br><br>

<input type="submit" value="Submit">

</form>

</body>
</html>
```

Discount.html

```
<!DOCTYPE html>
```

```
<html>
<head>
<title>Discount</title>
</head>

<body>

<h2>Select Discount</h2>

<form action="DiscountServlet" method="post">

<input type="radio" name="discount" value="0"> None<br>
<input type="radio" name="discount" value="5"> 5%<br>
<input type="radio" name="discount" value="10"> 10%<br>
<input type="radio" name="discount" value="15"> 15%<br><br>

<input type="submit" value="Calculate Bill">

</form>

</body>
</html>
```

ServiceServlet.java

```
import java.io.IOException;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

@WebServlet("/ServiceServlet")
public class ServiceServlet extends HttpServlet {
    private static final long serialVersionUID = 1L;

    protected void doGet(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException {
        String services[] = request.getParameterValues("service");
        HttpSession session = request.getSession();
        session.setAttribute("services", services);
    }
}
```

```
        response.sendRedirect("discount.html");}
        protected void doPost(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {
            doGet(request, response);
        }
    }
}
```

DiscountServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

@WebServlet("/DiscountServlet")
public class DiscountServlet extends HttpServlet {
    private static final long serialVersionUID = 1L;

    protected void doGet(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException {
        HttpSession session = request.getSession();

        String services[] = (String[]) session.getAttribute("services");

        String discountStr = request.getParameter("discount");

        session.setAttribute("discount", discountStr);

        int total = 0;

        PrintWriter out = response.getWriter();

        out.println("<html><body>");
```

```
        out.println("<h2>Selected Services</h2>");

        if(services != null)
        {
            for(String service : services)
            {
                if(service.equals("Face Spa"))
                {
                    out.println("Face Spa : ₹500<br>");
                    total += 500;
                }

                if(service.equals("Body Spa"))
                {
                    out.println("Body Spa : ₹1000<br>");
                    total += 1000;
                }

                if(service.equals("Foot Spa"))
                {
                    out.println("Foot Spa : ₹300<br>");
                    total += 300;
                }
            }
        }

        int discount = Integer.parseInt(discountStr);

        double discountAmount = total * discount / 100.0;

        double netAmount = total - discountAmount;

        out.println("<br>Total = ₹" + total);

        out.println("<br>Discount = " + discount + "%");

        out.println("<br>Net Payable = ₹" + netAmount);

        out.println("</body></html>");

    }
```

```
        protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
            doGet(request, response);
        }
    }
```

Output:

Select Spa Services

- ☒ Face Spa (â,¹500)
- ☒ Body Spa (â,¹1000)
- ☐ Foot Spa (â,¹300)

Submit

Select Discount

- ☐ None
- ☐ 5%
- ☐ 10%
- ☒ 15%

Calculate Bill

Selected Services

Face Spa : ₹500

Body Spa : ₹1000

Total = ₹1500

Discount = 15%

Net Payable = ₹1275.0

5.3. Create a project for FilmWare awards as follows:

HTML accepts the name of Best Movie(out of four options)should allow selection of Best Actor Male(out of Four options), next html form should allow selection of Best Actor Female(out of four options) which would be stored in a session variable. The last page should

Code:

Movie.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Best Movie</title>
</head>
<body>
  <h2>Select Best Movie</h2>

  <form action="ActorMaleServlet" method="post">
    <input type="radio" name="movie" value="RRR"> RRR <br>
    <input type="radio" name="movie" value="Jawan"> Jawan <br>
    <input type="radio" name="movie" value="Pathaan"> Pathaan <br>
    <input type="radio" name="movie" value="Animal"> Animal <br><br>

    <input type="submit" value="Next">
  </form>
</body>
</html>
```

ActorMaleServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/ActorMaleServlet")
public class ActorMaleServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        String movie = request.getParameter("movie");

        HttpSession session = request.getSession();
        session.setAttribute("movie", movie);

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        out.println("<h2>Select Best Actor (Male)</h2>");
        out.println("<form action='ActorFemaleServlet' method='post'>");

        out.println("<input type='radio' name='male' value='Shah Rukh Khan'> Shah Rukh  
Khan <br>");
        out.println("<input type='radio' name='male' value='Ranbir Kapoor'> Ranbir Kapoor  
<br>");
        out.println("<input type='radio' name='male' value='Ranveer Singh'> Ranveer Singh  
<br>");
        out.println("<input type='radio' name='male' value='Hrithik Roshan'> Hrithik  
Roshan <br><br>");

        out.println("<input type='submit' value='Next'>");
        out.println("</form>");
    }
}
```

ActorFemaleServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/ActorFemaleServlet")
public class ActorFemaleServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        String male = request.getParameter("male");

        HttpSession session = request.getSession();
        session.setAttribute("male", male);

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        out.println("<h2>Select Best Actor (Female)</h2>");
        out.println("<form action='ResultServlet' method='post'>");

        out.println("<input type='radio' name='female' value='Alia Bhatt'> Alia Bhatt  
<br>");
        out.println("<input type='radio' name='female' value='Deepika Padukone'> Deepika  
Padukone <br>");
        out.println("<input type='radio' name='female' value='Katrina Kaif'> Katrina Kaif  
<br>");
        out.println("<input type='radio' name='female' value='Kriti Sanon'> Kriti Sanon  
<br><br>");

        out.println("<input type='submit' value='Show Result'>");
        out.println("</form>");
    }
}
```

ResultServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/ResultServlet")
public class ResultServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        String female = request.getParameter("female");

        HttpSession session = request.getSession();
        session.setAttribute("female", female);

        String movie = (String) session.getAttribute("movie");
        String male = (String) session.getAttribute("male");

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        out.println("<h2> Filmfare Awards Result</h2>");
        out.println("Best Movie: " + movie + "<br>");
        out.println("Best Actor (Male): " + male + "<br>");
        out.println("Best Actor (Female): " + female + "<br>");
    }
}
```

Output:

Select Best Movie

- ☐ RRR
- ☐ Jawan
- ☐ Pathaan
- ☒ Animal

Next

Select Best Actor (Male)

- ☒ Shah Rukh Khan
- ☐ Ranbir Kapoor
- ☐ Ranveer Singh
- ☐ Hrithik Roshan

Next

Select Best Actor (Female)

- ☒ Alia Bhatt
- ☐ Deepika Padukone
- ☐ Katrina Kaif
- ☐ Kriti Sanon

Show Result

Conclusion

In this experiment, we successfully implemented **session tracking using HttpSession in Java Servlets**. We learned how to store user data using `setAttribute()` and retrieve it using `getAttribute()` across multiple web pages.

Through different applications like **Username storage, Raj Spa service selection, and Filmfare Awards voting**, we understood how session management helps maintain user state in a web application. It ensures smooth data flow between multiple requests without losing user input.

Thus, **HttpSession is an effective and widely used technique** for managing sessions in web applications such as login systems, shopping carts, and multi-step forms.